

4 (a) Define resistance.

.....
..... [1]

(b) Figure 4.1 shows the variation with temperature of the resistivity of copper.

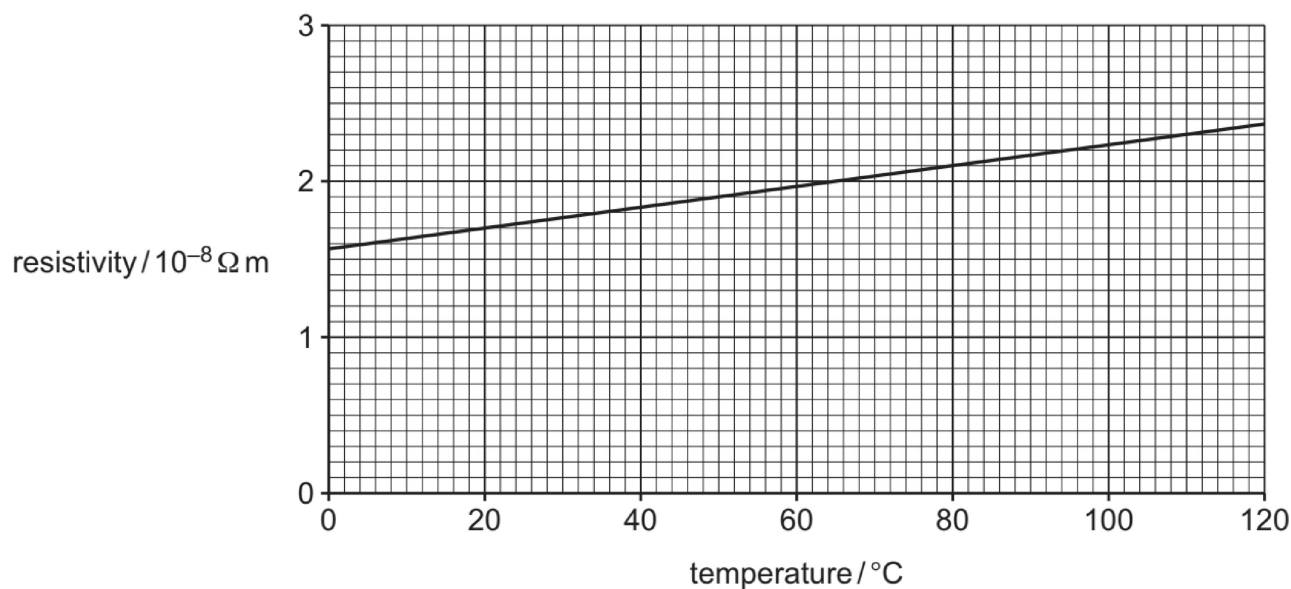


Figure 4.1

A copper wire has a length of 2.0 m and a diameter of 0.60 mm.

(i) Show that the cross-sectional area of the wire is $2.8 \times 10^{-7} \text{ m}^2$.

[1]

(ii) Determine the resistance of the wire at 20°C .

resistance = Ω [2]

- (iii) A different copper wire has the same resistance as the answer in 4(b)(ii). This wire also has a diameter of 0.60 mm but has a length that is greater than 2.0 m.

With reference to Figure 4.1, state and explain whether the temperature of this wire is less than, equal to or greater than 20 °C.

.....

 [2]

- (iv) State **one** way in which the variation of resistance with temperature for a thermistor is different from that of a copper wire.

..... [1]

- (c) Each metal has a property known as its temperature coefficient of resistivity α . This is given by

$$\alpha = \frac{\text{change in resistivity}}{\text{initial resistivity} \times \text{change in temperature}}.$$

- (i) Deduce the SI unit for α .

unit [1]

- (ii) Use Figure 4.1 to determine a value of α , in its SI unit, for copper at a temperature of 0 °C.

$\alpha =$ SI unit [2]

(d) There is a current of 5.6A in the wire in **4(b)**.

The number density of charge carriers (free electrons) for copper is $8.5 \times 10^{28} \text{ m}^{-3}$.

Calculate the mean drift speed v of the charge carriers in the wire.

$v = \dots\dots\dots \text{ m s}^{-1}$ [2]

[Total: 12]